

CHEMISTRY GRADE-XI

Revision Question Set

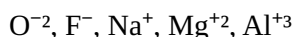
Physical Chemistry

1. What is universal gas constant and calculate the value of R in $\text{L. atm mol}^{-1}\text{K}^{-1}$. The volume of carbon monoxide gas collected over water at 25°C is 680 cc with a total pressure of 752 mm Hg. The vapour pressure of water at 25°C is 23.8 mm Hg. Determine the partial pressure of CO in container.
2. Sketch the diagram of P against $1/V$ and PV against V at constant temperature for verification of Boyle's law graphically. Calculate the mass of oxygen gas whose volume is 320 ml. at 17°C and 2 atm pressure.
3. State Boyle's law and mention its important application. Derive $PV=nRT$ with their usual meaning. A flask of 0.3 L capacity was weighed after it had been evacuated. It was then filled with a gas of unknown molecular mass at 1.0 atm and temperature of 300K the increase mass of flask was 0.977 g. calculate the molecular mass of the gas.
4. State and explain Graham's law of diffusion. A saturated hydrocarbon having molecular formula $\text{C}_n\text{H}_{2n+2}$ diffuses through a porous membrane twice as fast as Sulphur dioxide. Calculate the volume occupied by the hydrocarbon at 27°C and 2 atm pressure.
5. How is ideal gas differ from real gas? A vessel contains 12g of an ideal gas at $t^\circ\text{C}$ and 1 atm pressure. When the temperature is increased by 10°C at the same volume, the pressure is increased by 10%. Calculate the volume and initial temperature (molecular mass of gas=120).
6. Derive an ideal equation $PV=nRT$ where symbol have their usual meaning. A hydrocarbon C_xH_y has mass ratio between hydrogen and carbon is 1:10.5. one litre of the hydrocarbon at 127°C and 1 atmospheric pressure weights 2.8gm, find the molecular formula of the hydrocarbon.
7. How did Charle's law give the concept of absolute scale temperature? Plot temperature-volume relationship indicating absolute zero. Write its important application. A spherical balloon of 21cm diameter is to be filled with hydrogen gas at NTP from cylinder containing gas at 20 atms and 27°C . If the cylinder can hold 2.82 litres of water at NTP, Calculate the number of balloons that can be filled.
8. State and explain Charle's law. A vessel contains 12g of an ideal gas at $t^\circ\text{C}$ temperature and 1 atm pressure. When the temperature is increased by 10°C at the same volume, the pressure increased by 10%. Calculate the volume and initial temperature. (V.P. density of gas = 60).
9. What are the basic postulates of kinetic theory of gas? An evacuated glass vessel weighs 50gm when empty, 148gm when filled with a liquid of density of 0.98 gm/cc and 50.5gm when filled an ideal gas at 760mm of Hg and at 300K. Determine the molecular mass of the gas.
10. Grahams law of diffusion or effusion was formulated by Scottish physical chemist Thomas Graham in 1848. This law describes the relation between density of gas and rate of diffusion.
 - i. State the Graham's law of diffusion.
 - ii. How diffusion is differ from effusion.

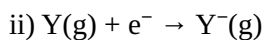
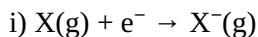
- iii. How is Grahams law used in everyday life?
 - iv. What volume of nitric oxide will diffuse out of a porous partition in 10 seconds when 25 ml of methane diffuse out in 4 seconds?
- 11.** Define universal gas constant with significances. How did Charle's law give the concept of absolute scale temperature? Plot temperature-volume relationship indicating absolute zero. The rate of diffusion of a saturated hydrocarbon (C_nH_{2n+2}) gas is 1.206 times that of SO_2 gas under identical conditions. Find the molecular mass and the molecular formula of this gas.
 - 12.** Define the following terms:
a) Evaporation b) Condensation c) Boiling point d) Vapour pressure e) Viscosity f) Surface tension
 - 13.** Write any two differences between:
a) crystalline solid and amorphous solid b) isotropic substance and anisotropic substance c) evaporation and boiling point
 - 14.** Define the following terms with suitable examples:
a) deliquescent substances b) efflorescent substances c) hygroscopic substances d) water of crystallization
 - 15.** Define liquid crystal and their types with suitable examples. Mention it's any four application.
 - 16.** Explain why?
a) Liquid drops are spherical in shape b) Water forms concave and mercury form convex meniscus in the glass tube.
c) glycerin is more viscous than water d) Gases do not settle on the bottom of container.
e) Evaporation takes place from the surface of liquid. f) Vegetables are cooked with the difficulty at a higher altitude.
 - 17.** Under what condition, real gas deviate from ideal behavior and why? A straight glass tube of length of 25cm has two inlet (A) and (B). Ammonia gas through inlet A and HCl gas through inlet B are allowed to enter the tube at the same time. White fumes of ammonium chloride is appeared at a point M inside the tube. Find the distance of point M from inlet A.
 - 18.** State and explain Dalton's law of partial pressure. When 2g of a gas (A) is introduced into a evacuated flask kept at $25^\circ C$, the pressure is found to be one atmosphere. If 3g of another gas (B) is added to the same flask, the total pressure becomes 1.5 atm at the same temperature. Assuming ideal behavior of gases, calculate the ratio of molecular mass.
 - 19.** Define efflorescent solids with example. A spherical balloon of 21 cm diameter is to be filled with hydrogen gas at NTP from a cylinder containing gas at 20 atm and $27^\circ C$. If the cylinder can hold 2.82 litres of water at NTP, calculate the number of balloon that can be filled up.
 - 20.** Define surface tension of liquid. How is surface tension of a liquid originated? Two vessels of capacity 1.5 litres and 2 litres contain hydrogen gas and oxygen gas respectively under a pressure of 750 mm and 100mm. The gases are mixed together in a vessel. What will be the final pressure of mixture?

Inorganic Chemistry

1. Justify that following species are isoelectronic. Compare the size of these species with appropriate reasoning and write them in the order of increasing size.



2. You have two reactions associated with the elements X and Y:



A certain amount of energy is released in both reactions. If addition of an electron to the valence shell of Y(g) was easier than to the valence shell of X(g), then which reaction, reaction (i) or reaction (ii), would have more negative number for energy change? Why? What is this energy change called?

3. Arrange the following elements in the order of their increasing metallic character. Justify answers.

a. K, Rb, Li b. Na, Mg, Al c. K, Ca, Rb

4. Define ionization energy of an element. How does it vary in the periodic table? What are the factors affecting ionization energy of an element?

5. How do you account for the following facts?

a. IE_1 of magnesium is more than that of aluminium. b. IE_1 of phosphorous is more than that of sulphur.

6. Define electronegativity of an element. How does it vary in the periodic table? Name the most electronegative and the least electronegative element.

7. Explain:

a. A cation is smaller than its parent atom. b. An anion is larger than its parent atom. c. Noble gas has zero electron affinity. d. Halogen has the highest electron affinity in the period. e. IE_2 of sodium is very high than its IE_1 .

8. Arrange the following ions in their increasing order of size, with explanations:

a. Na^{+} , Mg^{2+} , Al^{3+} , Si^{4+} b. P^{3-} , S^{2-} , Cl^{-} , Ar c. Na^{+} , K^{+} , Rb^{+} , Cs^{+} d. Cl^{-} , Br^{-} , I^{-} e. I, I^{-} , I^{+}

9. Compare the sizes (with explanations) of:

a. Li and C b. F and Cl c. K^{+} and Ca^{2+} d. F^{-} and F e. S^{2-} and Cl^{-}

10. Define IE. How do nuclear charge and size of the atom influence it? Arrange the following elements in the decreasing order of their ionization energies.

a. F, Cl, Br, I b. Li, Na, K, Rb

11. Write a reaction that shows nascent hydrogen is a more powerful reducing agent than molecular hydrogen. List the uses of different isotopes of hydrogen?

12. Oxygen is third most abundant element by mass which readily forms oxides with other elements. Some of the oxides are given below.

Na_2O	Al_2O_3	CO	SO_2	Fe_2O_3	H_2O_2
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- i) Identify the acidic oxide, basic oxide, neutral oxide and mixed oxide from the above table.
 - ii) Write two chemical equations to prove that the particular oxide is amphoteric in nature.
 - iii) Why is CO a harmful gas? iv) Write any one industrial applications of oxygen gas.
13. What are oxides? Classify the following oxides with justifications.
a. MnO b. BaO c. BaO₂ d. ZnO e. Fe₂O₃ f. KO₂ g. Al₂O₃
 14. Write short notes on: oxides of metals and non-metals. Why is F₂O₂ is not considered as oxide?
 15. Illustrate that H₂O₂ is green oxidant in chemical synthesis. N₂O₅ is acidic but NO is neutral. Why?
 16. Write the resonance structures of ozone molecule. Discuss the causes and harmful effects of ozone layer depletion?
 17. How can ozone be prepared artificially? What are the adverse effects of ozone layer depletion?
 18. Write short notes on:
a) Basic nature of ammonia b) Oxidizing nature of nitric acid
 19. What are the actions of:
a. Conc. Nitric acid upon iron b. dilute nitric acid upon magnesium
c. Dilute nitric acid upon copper d. conc. Nitric acid upon SO₂
 20. Write short note on ring test of nitrate. Why does ammonia turn mercuric nitrate paper into black?

Organic Chemistry

1. What is mean by catenation? Why are organic compounds more in number than inorganic compounds?
2. Define organic chemistry and state the difference between organic and inorganic compounds.
3. How are organic compounds classified? Give examples.
4. What are homocyclic and heterocyclic compounds? Give one example of each.
5. What is meant by functional group? Name five functional groups that contains oxygen atoms.
6. Write down the structural, condensed and bond line formula of following organic molecules;
a. Ethanol b. ethanal c. butanoic acid d. pentene
7. What is cracking? A fuel has octane number 80, what does it mean?
8. What is homologous series? Write its characteristics.
9. Write the homologous series of carboxylic acid and amine.
10. Write short notes on
a. Catenation property b. Tetravalency of carbon c. octane number d. aromatization

- 11.** Cholesterol is an organic compound. It is essential structural component of all animal cell membrane. However, if its level in the blood is high, it may lead to the risk of heart attack.
- What are the functional groups in cholesterol?
 - Give the molecular formula of cholesterol.
 - Is this alicyclic or aromatic compound? Explain.
- 12.** Write the functional group present in the following compound;
- Vanilline
 - Capsaicin
 - Cholesterol
- 13.** Define foreign element. How can you detect the foreign elements nitrogen, sulphur, halogen in given organic compound? Write the reaction.